Groups and Lie Algebras: Exercise Sheet 2

Exercise 2.1: Simple groups

A group G is called **simple** if it has exactly two normal subgroups: the trivial subgroup $\{1\}$ and G itself.

a) Let G be a simple group and H a group. Show that every group homomorphism from G to H is either an injection or the constant map at the identity of H. Show that any group G' satisfying this property must be simple.

Answer:

> Let $f : G \to H$ be a group homomorphism. Then by the isomophism theorem the kernel if f is a normal subgroup of G, i.e. $\ker f \in \{\{1\}, G\}$. If $\ker f = G$ then f is the constant map at the identity of H. To show that if $\ker f = \{1\}$ impliels f is an injection consider $g_1, g_2 \in G$ such that $f(g_1) = f(g_2)$. Then $f(g_2^{-1}g_1) = f(1) = 1$ and thus $g_2^{-1}g_1 \in \ker f = \{1\}$ which implies $g_1 = g_2$.
>
> Conversely, let G' be a group such that every group homomorphism $h : G' \to H$ for any group H is either an injection or the constant map at the identity of H. Let $N \leq G'$ be a normal subgroup. Then there exists a group homomorphism f_1 such that $\ker f = N$. If f_1 is an injection then $\ker f = \{1\}$, and if f_1 is the constant map at the identity of H then $\ker f = G'$. In either case $N \in \{\{1\}, G'\}$ and thus G' is simple. ■

b) For which $n \in \mathbb{N}$ is $\mathbb{Z}/n\mathbb{Z}$ simple?

Answer:

> Let $N \leq \mathbb{Z}/n\mathbb{Z}$ be a normal subgroup. Since $\mathbb{Z}/n\mathbb{Z}$ is abelian, every subgroup is normal. Since every subgroup of $\mathbb{Z}/n\mathbb{Z}$ is of the form $k\mathbb{Z}/n\mathbb{Z}$ for some $k \in \mathbb{N}$ dividing n. Therefore, $\mathbb{Z}/n\mathbb{Z}$ is simple if and only if n has no divisors other than 1 and n, i.e. if and only if n is prime. ■

c) Let G be a group and $N \subset G$ a maximal normal subgroup, i.e. $N \neq G$ and

$$N \subsetneq N' \subset G$$

with N' normal implies $N' = G$. Show that G/N is simple.

Answer:

> Consider the canonical projection $\pi : G \to G/N, \quad g \mapsto [g]$ and $H \trianglelefteq G/N$. Then the preimage $\pi^{-1}(H)$ is a normal subgroup of G since for $g \in G$ and $h \in \pi^{-1}(H)$ we have $\pi(ghg^{-1}) = \pi(g)\pi(h)\pi(g^{-1}) \in H$ since H is normal. But $N \subset \pi^{-1}(H)$ since $\pi^{-1}(1) = N$. Thus either $\pi^{-1}(H) = N$ or $\pi^{-1}(H) = G$ by the maximality of N. In the first case $H = 1$ and in the second case $H = G/N$. Therefore, G/N is simple. ■

d) Show that a group with only two conjugacy classes is simple.

Answer:

First we show that every normal subgroup $N \trianglelefteq G$ is a union of conjugacy classes. Let $h \in N$, since N is normal we have $ghg^{-1} \in N \, \forall g \in G$, i.e. the conjugacy class of h is contained in N. Therefore, N is the union of the conjugacy classes of its elements.

For every group $[1] = \{1\} \in \mathrm{conj}(G)$. If G has only two conjugacy classes, then the second conjugacy class is $G/\{1\}$. Every normal subgroup $N \trianglelefteq G$ is a union of conjugacy classes, so either $N \in \{\{1\}, G/\{1\}, G\}$. But $G/\{1\}$ is not a subgroup of G since it does not contain the identity, so $N \in \{\{1\}, G\}$ and thus G is simple. ■

e) Iterating the construction from c), we can assign to each finite group G a sequence of simple groups. Show explicitly for $G = \mathbb{Z}/12\mathbb{Z}$ that the simple groups appearing in this decomposition are independent of the choice of maximal normal subgroup (this is true for every finite group).

Answer:

To iterate the construction we start with $G \trianglerighteq G_1$ where G_1 is a maximal normal subgroup of G. Then G/G_1 is simple and therefore of prime order. This is iterated by choosing a maximal normal subgroup G_2 of G_1 and so on until we reach the trivial group. The simple groups appearing in this decomposition are the quotients $G/G_1, G_1/G_2, \ldots$. We show that for $G = \mathbb{Z}/12\mathbb{Z}$ the simple groups appearing in this decomposition are independent of the choice of maximal normal subgroup. Since G is abelian, every subgroup is normal. For $\mathbb{Z}/k\mathbb{Z}$ to be a subgroup of $\mathbb{Z}/12\mathbb{Z}$ we need k to divide 12. The divisors of 12 are $1, 2, 3, 4, 6, 12$. For a subgroup to be maximal k needs to be a proper divisor of 12 such that there exists no proper divisor of 12 which is a multiple of k. The maximal normal subgroups of $\mathbb{Z}/12\mathbb{Z}$ are therefore $\mathbb{Z}/4\mathbb{Z}$ and $\mathbb{Z}/6\mathbb{Z}$.

Starting with $G_1 = \mathbb{Z}/4\mathbb{Z}$ we have $G/G_1 \cong \mathbb{Z}/3\mathbb{Z}$. The maximal proper divisor of 4 is 2 and thus $G_2 = \mathbb{Z}/2\mathbb{Z}$ and therefore the second simple group is $G_1/G_2 \cong \mathbb{Z}/2\mathbb{Z}$. Finally, the maximal proper divisor of 2 is 1 and thus $G_3 = \{0\}$ and the third simple group is $G_2/G_3 \cong \mathbb{Z}/2\mathbb{Z}$. Therefore the decomposition yields the simple groups $\mathbb{Z}/3\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}$.

On the other hand, starting with $G_1 = \mathbb{Z}/6\mathbb{Z}$ we have $G/G_1 \cong \mathbb{Z}/2\mathbb{Z}$. The maximal proper divisors of 6 are $2, 3$ and thus $G_2 = \mathbb{Z}/3\mathbb{Z}$ or $G_2 = \mathbb{Z}/2\mathbb{Z}$. Therefore the second simple group is $G_1/G_2 \cong \mathbb{Z}/2\mathbb{Z}$ or $G_1/G_2 \cong \mathbb{Z}/3\mathbb{Z}$. Finally, the maximal proper divisor of 3 is 1 as it is of 2 and thus $G_3 = \{0\}$ for both cases and the third simple group is $G_2/G_3 \cong \mathbb{Z}/3\mathbb{Z}$ or $G_2/G_3 \cong \mathbb{Z}/2\mathbb{Z}$. Therefore the decomposition yields the simple groups $\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/3\mathbb{Z}$.

Thus all possible decompositions of $\mathbb{Z}/12\mathbb{Z}$ yield the same simple groups $\mathbb{Z}/3\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}$. ■

f) The decomposition in e) is called the Jordan-Hölder decomposition of G. Show that there are non-isomorphic groups with the same Jordan-Hölder decomposition.

Answer:

Exercise 2.2: Spontaneous symmetry breaking

It can happen that a quantum mechanical system is in a ground state which does not preserve the symmetries of the system. In this case we say the symmetry is *spontaneously broken*.
Consider a particle on a circle with Hilbert space

$$\mathcal{H} = L^2(\mathbb{R}/\mathbb{Z})$$

and Hamiltonian

$$\mathbb{H} = \frac{1}{2}\left(-i\partial_x - \frac{1}{2}\right)^2.$$

Compute the eigenfunctions of $\mathbb{H}$ and show that

$$f(x) \mapsto f(-x)$$

is a symmetry. What is the action of the symmetry when applied to the eigenfunctions?

Answer:

To find the eigenfunctions of $\mathbb{H}$ we find the characteristic polynomial of $\mathbb{H}$. For $f \in \mathcal{H}$ we have

$$\mathbb{H}f = \frac{1}{2}\left(-i\partial_x - \frac{1}{2}\right)^2 f = \frac{1}{2}\left(-\partial_x^2 + i\partial_x + \frac{1}{4}\right) f = Ef$$

therefore the characteristic polynomial of $\mathbb{H}$ with the ansatz $f(x) = e^{\lambda x}$ is given by

$$-\lambda^2 + i\lambda + \frac{1}{4} - 2E = 0$$

The solutions of this polynomial are given by the pq formula

$$\lambda = \frac{i}{2} \pm \sqrt{-2E} = i\left(\frac{1}{2} \pm \sqrt{2E}\right)$$

and thus the eigenfunctions of $\mathbb{H}$ are given by

$$f(x) = Ae^{i\left(\frac{1}{2}+\sqrt{2E}\right)x} + Be^{i\left(\frac{1}{2}-\sqrt{2E}\right)x}$$

for some $A, B \in \mathbb{C}$. Since f is takes values on the circle, we have $f(x) = f(x + 2\pi)$ and thus

$$\begin{aligned} Ae^{i\left(\frac{1}{2}+\sqrt{2E}\right)x} + Be^{i\left(\frac{1}{2}-\sqrt{2E}\right)x} &= Ae^{i\left(\frac{1}{2}+\sqrt{2E}\right)(x+2\pi)} + Be^{i\left(\frac{1}{2}-\sqrt{2E}\right)(x+2\pi)} \\ &= Ae^{i\left(\frac{1}{2}+\sqrt{2E}\right)x}e^{i\left(\frac{1}{2}+\sqrt{2E}\right)2\pi} + Be^{i\left(\frac{1}{2}-\sqrt{2E}\right)x}e^{i\left(\frac{1}{2}-\sqrt{2E}\right)2\pi} \end{aligned}$$

and thus

$$e^{i\left(\frac{1}{2}+\sqrt{2E}\right)2\pi} = e^{i\left(\frac{1}{2}-\sqrt{2E}\right)2\pi} = 1$$

which implies $\frac{1}{2} \pm \sqrt{2E} \in \mathbb{Z}$ and thus $E = \frac{1}{2}\left(n \mp \frac{1}{2}\right)^2$ for some $n \in \mathbb{Z}$. Therefore, the eigenfunctions of $\mathbb{H}$ are given by

$$f(x) = Ae^{inx} + Be^{-inx}$$

for some $A, B \in \mathbb{C}$ and $n \in \mathbb{Z}$. The eigenfunctions of $\mathbb{H}$ transform under parity as

$$f(-x) = Ae^{-inx} + Be^{inx}$$

which is still an eigenfunction of $\mathbb{H}$ with the same eigenvalue E. Therefore, the parity transformation is a symmetry of the system. ■

Exercise 2.3: Symmetries of the Dodecahedron

In this exercise we study the symmetry group of the dodecahedron.

a) Compute the size of the group of symmetries of the dodecahedron using the orbit-stabilizer theorem.

Answer:

Concider the faces of the dodecahedron. The orbit of a face are all the faces of the dodecahedron. Therefore $\mid \mathcal{O}(F) \mid = 12$. The stabilizer of a face is the symmetries of a pentagon, which consists of 5 rotations and 5 reflections, and thus $\mid \mathrm{Stab}(F) \mid = 10$. Therefore, by the orbit-stabilizer theorem, the size of the group of symmetries of the dodecahedron is given by $\mid \mathrm{Sym}(\text{dodecahedron}) \mid = \mid \mathcal{O}(F) \mid \cdot \mid \mathrm{Stab}(F) \mid = 12 \cdot 10 = 120$. ■

b) What is the size of the group of rotational symmetries?

Answer:

The group of rotations is the subgroup which does not contain reflections. Since the group of symmetries of the dodecahedron contains 5 reflections for each face, the group of rotations contains 5 rotations for each face. Therefore, the size of the group of rotational symmetries is given by $\mid \mathrm{Rot}(\text{dodecahedron}) \mid = 12 \cdot 5 = 60$. ■

c) There are 5 cubes inside the dodecahedron which can be rotated into each other and contain all its vertices. Show that there is an injection from the group of rotational symmetries to S_5.

Answer:

To inscribe a cube into a dodecahedron, notice that the number of edges on a cube and the number of faces on a dodecahedron is 12. A cube is uniquely defined by a diagonal on a face of the dodecahedron. Since each face has 5 vertices there are 5 distinct cubes. Laber this cubes by $\mathcal{C} = \{\mathcal{C}_1, \cdots, \mathcal{C}_5\}$. The group of rotation preserves the order of vertices and thus the cubes. Therefore, there is a group action

of the rotational group on the set of cubes:

$$\mathrm{Rot}_{12} \times \mathcal{C} \to \mathcal{C}, \quad (r, \mathcal{C}_i) \mapsto r \triangleright \mathcal{C}_i$$

This induces a bijection $\sigma(g) : \mathcal{C} \to \mathcal{C}$ which is isomorphic to a permutation of the set $\{1, \cdots, 5\}$. Define the map

$$\phi : \mathrm{Rot}_{12} \to S_5, \quad g \mapsto \sigma(g)$$

ϕ is a homomorphism since for $g_1, g_2 \in \mathrm{Rot}_{12}$ and $\mathcal{C}_i \in \mathcal{C}$ we have

$$\begin{aligned}
\phi(g_1 g_2)(\mathcal{C}_i) &= \sigma(g_1 g_2)(\mathcal{C}_i) \\
&= (g_1 g_2) \triangleright \mathcal{C}_i \\
&= \sigma(g_1)(\sigma(g_2)(\mathcal{C}_i)) \\
&= (\sigma(g_1) \circ \sigma(g_2))(\mathcal{C}_i) \\
&= (\phi(g_1) \circ \phi(g_2))(\mathcal{C}_i)
\end{aligned}$$

ϕ is injective since for $g_1, g_2 \in \mathrm{Rot}_{12}$ such that $\phi(g_1) = \phi(g_2)$ we have:

$$\begin{aligned}
\phi(g_1)(\mathcal{C}_i) &= \phi(g_2)(\mathcal{C}_i) \\
\sigma(g_1)(\mathcal{C}_i) &= \sigma(g_2)(\mathcal{C}_i) \\
g_1 \triangleright \mathcal{C}_i &= g_2 \triangleright \mathcal{C}_i \\
g_2^{-1} g_1 \triangleright \mathcal{C}_i &= \mathcal{C}_i
\end{aligned}$$

and therefore $g_2^{-1} g_1$ is the identity. Thus $g_1 = g_2$ and ϕ is injective. ■

d) Show that the image of the injection from c) is contained in the alternating group A_5.
Answer:

■

e) Show that the group of symmetries of the dodecahedron is $A_5 \times \mathbb{Z}_2$.
Answer:

■

Exercise 2.4: Cayley's theorem

Show that every finite group is isomorphic to a subgroup of a symmetric group.
Hint: Use the action of a group on itself by left multiplication.

Answer:

Let G be a finite group. The action of G on itself by left multiplication is given by $L_g[h] \equiv g \triangleright h = gh$. Define the map $\phi : G \to \mathrm{Sym}_G, \quad g \mapsto L_g$.

First we show that this is well defined, i.e. that L_g is a bijective homomorphism for

every $g \in G$.

- L_g is a homomorphism: For $h_1, h_2 \in G$ we have $L_g[h_1h_2] = g(h_1h_2) = (gh_1)(gh_2) = L_g[h_1]L_g[h_2]$.
- L_g is injective: For $h_1, h_2 \in G$ such that $L_g[h_1] = L_g[h_2]$ we have $gh_1 = gh_2$ and thus $h_1 = h_2$.
- L_g is surjective: For $h \in G$ we have $L_g[g^{-1}h] = g(g^{-1}h) = h$.

Next we show that ϕ is a homomorphism. For $g_1, g_2, h \in G$ we have

$$\phi(g_1g_2)[h] = L_{g_1g_2}[h] = (g_1g_2)h = g_1(g_2h) = L_{g_1}[L_{g_2}[h]] = (\phi(g_1) \circ \phi(g_2))[h]$$

Finally to show that ϕ is an isomorphism we show that it is injective. For $g_1, g_2 \in G$ such that $\phi(g_1) = \phi(g_2)$ we have $L_{g_1} = L_{g_2}$ and thus $g_1h = g_2h$ for every $h \in G$. In particular, for $h = 1$ we have $g_1 = g_2$. Therefore, ϕ is an injection and thus G is isomorphic to a subgroup of Sym_G. ∎